

CLINICAL PROBLEM-SOLVING

Caren G. Solomon, M.D., M.P.H., *Editor*

Corycat

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In this Journal feature, information about a real patient is presented in stages (boldface type) to an expert clinician, who responds to the information by sharing relevant background and reasoning with the reader (regular type). The authors' commentary follows.

A 75-year-old man presented with a 4-day history of a pruritic rash on his lower legs. He noted generalized malaise and an unintentional weight loss of 7 kg over the preceding 2 months. He reported no fevers, headaches, cough, dyspnea, abdominal pain, or arthralgias but had an untreated hepatitis C virus infection. He took no medications. He reported occasional intranasal cocaine use but no history of alcohol or tobacco use. Until the previous month, he had lived in a rural location in New Mexico, where he had kept six cats as pets. At the time of presentation, he resided in a homeless shelter in San Francisco.

This patient presents with concerns that are acute: a pruritic rash on his lower legs and unintentional weight loss. He also has malaise. Potential causes of malaise and weight loss include malignant, infectious, and inflammatory processes and psychiatric disorders.

Hepatitis C may cause mixed cryoglobulinemia, in which patients often present with palpable purpura on the legs. The history of exposure to cats raises the possibility of infection with *Bartonella henselae*, which is the causative agent in cat scratch disease but in rare instances may also cause endocarditis. Exposure to cats and a lack of personal housing are also risk factors for murine typhus, in which patients may present with a rash early in the course of illness. Residence in a shelter is a risk factor for exposure to tuberculosis, and persons who are homeless are at higher risk for *B. quintana* infection, a relatively common cause of culture-negative endocarditis. Q fever, caused by *Coxiella burnetii* infection, is also a cause of culture-negative endocarditis and would be a consideration if the patient has had exposure to farm animals in rural New Mexico.

The patient's temperature was 36.7°C, the blood pressure 115/54 mm Hg, the heart rate 61 beats per minute, and the respiratory rate 18 breaths per minute. The oxygen saturation was 99% while the patient was breathing ambient air. He appeared comfortable. There were no oral ulcers or exudates in his oropharynx, and there was no lymphadenopathy. The jugular venous pulsation was normal. He had a new 3/6 systolic murmur that was best heard at the right upper sternal border. His lungs were clear to auscultation. The abdomen was nontender, with no hepatosplenomegaly. No synovitis was noted on musculoskeletal examination. There were coalescing purple macules and patches on the patient's legs that did not blanch with pressure, and he had palpable purpura on his thighs (Fig. 1). No abnormalities were observed on neurologic examination.

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Figure 1. Photograph of the Patient's Thigh Showing Nonblanching Purpuric Papules and Macules.

Notable findings on physical examination include a systolic murmur and palpable purpura. The latter is a result of inflammation of the capillaries, the venule walls, or both, leading to extravasation of blood into the skin. Palpable purpura are a hallmark of cutaneous vasculitis. Causes include autoimmune disorders, cryoglobulinemia caused by infection or an autoimmune disorder, antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis, bacterial endocarditis, cancer, and certain medications, such as allopurinol.

Further characterization of the systolic murmur would be useful; infective endocarditis could explain a systolic murmur and a purpuric rash. Nonbacterial thrombotic endocarditis, which may occur in patients with systemic lupus erythematosus or cancer, could also account for the presence of both a systolic murmur and palpable purpura.

The white-cell count was 4500 per cubic millimeter, and the differential count was normal. The hemoglobin level was 11.6 g per deciliter (7.2 mmol per liter), with a mean corpuscular

volume of 102 fl, and the platelet count was 153,000 per cubic millimeter. The blood urea nitrogen level was 51 mg per deciliter (18 mmol per liter), and the serum creatinine level was 2.4 mg per deciliter (212 μ mol per liter). The chronicity of the abnormal creatinine level was unknown. The aspartate aminotransferase level was 96 U per liter (normal value, 10 to 48), the alanine aminotransferase level was 67 U per liter (normal value, 10 to 40), and the total bilirubin level was 1.9 mg per deciliter (32 mmol per liter). The erythrocyte sedimentation rate was 89 mm per hour (normal value, 0 to 19). The urinalysis showed 50 to 100 red cells and 5 to 10 white cells per high-power field. The urine protein:creatinine ratio was 35 (with protein measured in milligrams and creatinine in millimoles) (reference value, <8). Urine toxicologic testing was positive for cocaine. Renal ultrasonography showed bilateral echogenic kidneys of normal size, with no hydronephrosis. The hepatitis C viral load was 1.6 million IU per milliliter. Tests for antibodies to human immunodeficiency virus and hepatitis B surface antigen were negative. The complement C3 level was 42 mg per deciliter (normal value, 71 to 159), and the complement C4 level was 11 mg per deciliter (normal value, 13 to 30). Two sets of blood cultures were obtained. Findings on chest radiography were normal.

The presence of hematuria, proteinuria, low C3 and C4 levels, and an elevated creatinine level suggest immune-complex-mediated glomerulonephritis. Hypocomplementemic glomerulonephritis with a purpuric rash can occur in hepatitis C-associated cryoglobulinemia, infective endocarditis, or autoimmune disorders. Echocardiography can be used to evaluate valvular function, underlying valvular pathology, vegetations, and periannular abnormalities. If the cause of suspected glomerulonephritis is not apparent in the infectious or autoimmune work-up, renal biopsy may be considered.

Test results for antinuclear antibody and ANCA were negative. An anti-streptolysin O antibody titer was negative. The level of rheumatoid factor was 207 IU per milliliter (normal value, <14). The level of anti-Ro-52 antibody was 57 IU per milliliter (normal value, <40). A serum cryoglobulin test was negative. Blood cultures showed no growth at 5 days. A biopsy specimen of the skin

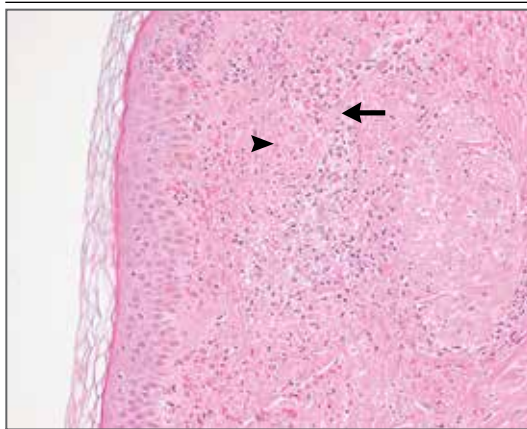


Figure 2. Photomicrograph of a Skin-Biopsy Specimen.

Staining with hematoxylin and eosin shows small-vessel vasculitis. A superficial blood vessel shows endothelial damage (arrow), and extravasated red cells can be seen in the dermis (arrowhead).

showed leukocytoclastic vasculitis of the small vessels of the papillary dermis (Fig. 2), with C3 (3+) and IgM (2+) and no IgA or IgG. No fibrinogen was observed on immunofluorescence. A transthoracic echocardiogram showed a left ventricular ejection fraction of 65 to 70%, mildly thickened aortic-valve leaflets, mild aortic-valve sclerosis, and moderate aortic regurgitation; no vegetations were observed (Fig. 3 and video, available with the full text of this article at NEJM.org). A renal biopsy showed an immune-complex-mediated glomerulonephritis (Fig. 4). Immunofluorescence showed staining of IgG, IgM, IgA, C3, C1q, fibrinogen, kappa, and lambda, with a predominance of IgA.

The biopsy specimen of the skin shows leukocytoclastic vasculitis, as was anticipated on the basis of the appearance of the rash and the concomitant presence of glomerulonephritis. The results of laboratory testing do not support the diagnosis of ANCA-associated vasculitis, and the results of the renal biopsy indicate an immune-complex-mediated glomerulonephritis rather than a pauci-immune pattern. Among the entities causing immune-complex-mediated glomerulonephritis, post-streptococcal glomerulonephritis and lupus nephritis are unlikely on the basis of the laboratory test results and the absence of suggestive clinical features. Isolated elevation of anti-Ro-52 antibody and rheumatoid factor levels can be seen in a variety of nonautoimmune



Figure 3. Transthoracic Echocardiogram.

Moderate aortic regurgitation can be seen on color Doppler. No vegetations were observed.

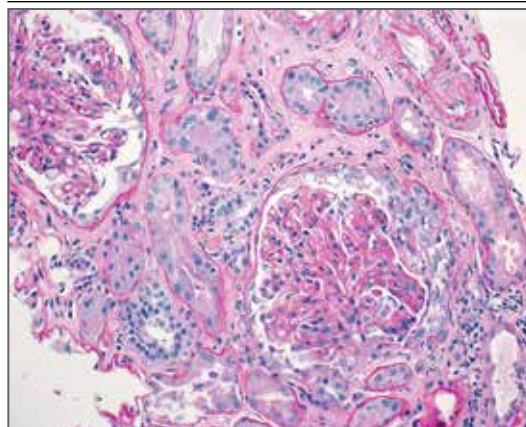


Figure 4. Renal-Biopsy Specimen.

Focal crescentic glomerulonephritis is revealed on periodic acid-Schiff staining.

disorders, including cancer, endocarditis, and chronic hepatitis C virus infection. Tests for serum cryoglobulins were negative, but false negative results can occur if testing is not performed properly. Specimens should be obtained with the use of warmed syringes and tubes and transported to the laboratory in a warm-water bath; if the sample temperature falls below 37°C, premature precipitation of cryoglobulins will occur. Patients with IgA vasculitis may present with purpura and glomerulonephritis, but this condition is much more common in children, and assessment of the skin-biopsy specimen under direct immunofluorescence does not show IgA, which would be expected in IgA vasculitis. Other forms of infectious and autoimmune-mediated small-vessel vasculitis may also be associated with glomerular deposition of IgA.



A video showing a transthoracic echocardiogram is available at [NEJM.org](https://www.nejm.org)

The transthoracic echocardiogram does not show a mobile vegetation, but its sensitivity is limited. A transesophageal echocardiogram should be obtained. Because blood cultures show no growth, entities causing culture-negative infective endocarditis should be considered. The patient has epidemiologic risk factors for infection with *B. quintana*, *B. henselae*, and *C. burnetii*, and bartonella and coxiella species are among the most common causes of culture-negative bacterial endocarditis. Serologic testing for bartonella species and *C. burnetii* should be performed. In patients with rapidly progressive glomerulonephritis, pulse doses of methylprednisolone are often given initially, but they are less likely to be beneficial if the underlying cause is current or previous infection.

The patient received 1000 mg of intravenous methylprednisolone daily for 3 days followed by 60 mg of oral prednisone per day for suspected hepatitis C–associated mixed cryoglobulinemic vasculitis. Sofosbuvir–velpatasvir was initiated for hepatitis C virus infection. The patient's kidney function improved, and at the time of discharge his serum creatinine level was 1.7 mg per deciliter (150 μ mol per liter). Two weeks later, rituximab was initiated as induction therapy for vasculitis. He reported that his rash was greatly diminished but that his malaise had worsened. Laboratory results showed a serum creatinine level of 3.4 mg per deciliter (301 μ mol per liter). He was readmitted to the hospital for worsening renal function.

Treatment with direct-acting antiviral agents is indicated in nearly all patients with active hepatitis C virus infection as evidenced by detection of hepatitis C virus RNA. In addition, antiviral treatment often addresses manifestations of hepatitis C–associated cryoglobulinemic syndromes.

The patient's clinical presentation, skin findings, and biopsy results are consistent with the suspected diagnosis of hepatitis C–associated mixed cryoglobulinemic vasculitis. The negative result for serum cryoglobulins does not rule out the diagnosis of a mixed cryoglobulinemia syndrome, particularly when clinical suspicion is high. The initial improvement in the patient's creatinine level and the substantial abatement of the rash suggest that he had a response to the antiviral therapy for hepatitis C, to the glucocor-

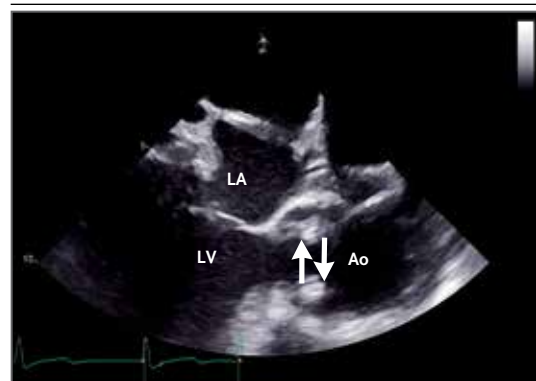


Figure 5. Transesophageal Echocardiogram Showing Calcified Aortic-Valve Leaflets, with Attached Vegetations.

Calcified leaflets with vegetations (arrows) suggest endocarditis. Ao denotes aorta, LA left atrium, and LV left ventricle.

ticoid therapy, or both, as is the case with the majority of patients with hepatitis C–associated mixed cryoglobulinemic vasculitis. However, given the recurrent deterioration in renal function, I would pursue further workup for other causes. A transesophageal echocardiogram should be obtained and serologic testing for bartonella and other potential culture-negative endocarditis pathogens should be performed.

The test result for *B. henselae* IgG was positive, with a titer of 1:262,144 (reference range for positivity, $\geq$ 1:256); that for *B. henselae* IgM was negative; that for *B. quintana* IgG was positive, with a titer of 1:512 (reference range for positivity, $\geq$ 1:256); and that for *B. quintana* IgM was negative. A transesophageal echocardiogram was obtained and showed aortic-valve vegetations (Fig. 5).

On the basis of the patient's extremely high IgG titer for *B. henselae* and the findings on transesophageal echocardiography, a diagnosis of bartonella endocarditis was made. As mentioned above, endocarditis can cause palpable purpura and hypocomplementemic glomerulonephritis, and the results of the biopsies of the skin and kidneys are also compatible with this diagnosis. The patient has a history of exposure to cats, which is a well-recognized risk factor for *B. henselae* endocarditis; his echocardiogram also suggests a preexisting aortic-valve abnormality. Bartonella endocarditis typically involves the aortic valve. Patients with bartonella endocarditis may

also present with ANCA-associated glomerulonephritis, although in this case the test result was negative. The modestly elevated level of IgG antibodies to *B. quintana* is probably the result of cross-reactivity with *B. henselae* antibodies.

Antibiotic therapy for bartonella endocarditis should be initiated. Although some experts recommend a combination of doxycycline plus gentamicin for initial treatment of bartonella endocarditis, gentamicin should be avoided in this patient given his impaired renal function. Rifampin, in combination with doxycycline, is a recommended alternative to gentamicin, but drug interactions with rifampin need to be considered. The patient had begun to receive sofosbuvir–velpatasvir for the treatment of hepatitis C; however, coadministration of rifampin with sofosbuvir–velpatasvir is contraindicated, since rifampin can have the effect of reducing levels of both sofosbuvir and velpatasvir.

Doxycycline and rifampin were initiated for the treatment of *B. henselae* endocarditis. Sofosbuvir–velpatasvir was withheld during antibiotic therapy owing to the potential for interaction with rifampin. Glucocorticoids were rapidly tapered. At follow-up 12 weeks after the initiation of antibiotic therapy, the patient reported that his malaise and rash had resolved. His serum creatinine level had declined to 1.4 mg per deciliter (124 μ mol per liter).

COMMENTARY

Bartonella species are gram-negative bacilli that cause a range of diseases in humans, including bacillary angiomatosis, cat scratch disease, bacteremia, and endocarditis. Of the more than 30 known bartonella species, *B. quintana*, transmitted by body lice, and *B. henselae*, transmitted by cats with claws that are contaminated with *B. henselae*-infected flea feces, cause most clinical infections.¹ Although serologic cross-reactivity occurs between *B. henselae* and *B. quintana*, the very high IgG antibody titer for *B. henselae* in this patient establishes this species as the causative pathogen. Infection with bartonella species is among the most common causes of culture-negative endocarditis in the United States.^{2,3}

Bartonella endocarditis can lead to vasculitic manifestations, such as glomerulonephritis. The renal-biopsy findings in patients with bartonella

endocarditis-associated glomerulonephritis are variable, but most case reports describe an immune-complex pattern of glomerular injury, sometimes with positive test results for ANCA.^{4,7} Less commonly, patients with bartonella endocarditis present with a pauci-immune glomerulonephritis and have positive test results for ANCA, findings that mimic primary ANCA-associated vasculitis.⁵ Immune-mediated complications of bartonella infection usually diminish with appropriate antibiotic therapy; in most cases, immunosuppression is not required.⁵

Randomized, controlled trials are lacking to guide the treatment of bartonella endocarditis. A 2003 retrospective report of a heterogeneous group of patients with bartonella endocarditis suggested that patients receiving at least 14 days of an aminoglycoside had improved outcomes.⁸ Although 2004 treatment guidelines⁹ were based on that report, it is prudent to avoid aminoglycosides in patients with associated glomerulonephritis or other conditions that may result in renal complications, given the potential for the development of irreversible renal failure. A regimen of rifampin (which has excellent intracellular penetration as well as bactericidal properties) plus doxycycline is recommended as an alternative⁹ and was effective in the treatment of the patient in this case. Because relapse is a risk in patients with endovascular and disseminated bartonella infections, the treatment of bartonella endocarditis should be prolonged. Bartonella antibody titers should be obtained every 4 to 8 weeks. Titers should decrease over a period of months.

Our patient presented with a murmur, palpable purpura, and hypocomplementemic glomerulonephritis, prompting consideration of diagnoses of endocarditis and small-vessel vasculitis. After the initial workup, in which blood cultures were negative but a high hepatitis C viral load was identified, cryoglobulinemic vasculitis was presumed to be the diagnosis. Cryoglobulins are cold-insoluble immunoglobulins that dissolve on rewarming. Type II and III cryoglobulins may be present in patients with chronic infections, such as hepatitis C, and are associated with glomerulonephritis and other vasculitic manifestations.

Two common cognitive processes engaged in by clinicians — heuristics and confirmation bias — probably contributed to the premature acceptance of a diagnosis of hepatitis C–associated

cryoglobulinemic vasculitis. Heuristics are mental shortcuts that all clinicians use to improve diagnostic efficiency. Unfortunately, these shortcuts can also lead to diagnostic errors.¹⁰ The constellation of leukocytoclastic vasculitis, immune-complex-mediated glomerulonephritis, a high titer of rheumatoid factor, and low complement levels in a patient with untreated hepatitis C fits the heuristic approach to the diagnosis of hepatitis C-associated cryoglobulinemic vasculitis. In complex cases, clinicians can reduce heuristic error by using cognitive forcing — the deliberate use of analytical thinking rather than pattern recognition. One approach to cognitive forcing is for the clinician to identify all reasonable alternative diagnoses and to then consider which can be definitively ruled out. In this case, bacterial endocarditis was an alternative diagnostic consideration, and transesophageal echocardiography was required to exclude it.

The second potential source of diagnostic error was confirmation bias — the overvaluing of evidence that supports an initial impression and the undervaluing of information that is in-

consistent with that impression.¹¹ In this case, overvaluing the absence of a mobile vegetation on transthoracic echocardiography and undervaluing the abnormal aortic valve are examples of confirmation bias. Contextual clues, such as the patient's exposure to cats and his heart murmur in conjunction with glomerulonephritis, led the treating physician to order serologic tests for bartonella species.

This case underscores the need to exclude the possibility of underlying endocarditis in cases of glomerulonephritis that are compatible with this diagnosis. Careful attention to risk factors such as zoonotic exposures and heart-valve pathology may raise suspicion for culture-negative endocarditis. In this case, the results of transesophageal echocardiography and serologic testing for bartonella species were key to diagnosing a copycat of hepatitis C-associated cryoglobulinemic vasculitis.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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